

EECS 195 LAB REPORT 5: Full Implementation of Motion Planning using ROS

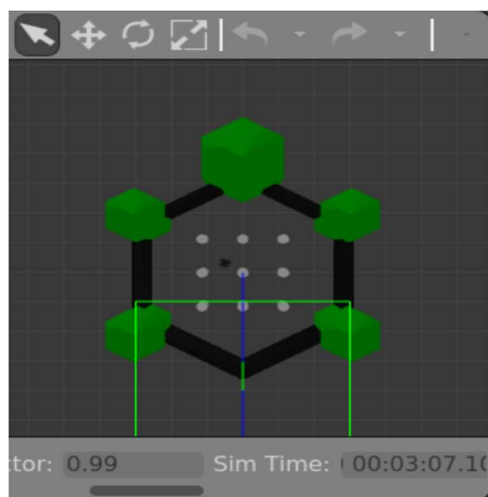
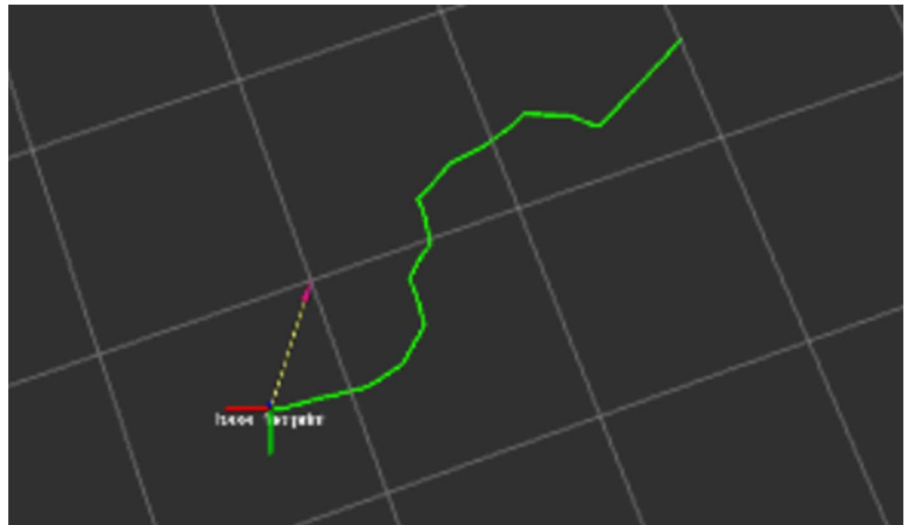
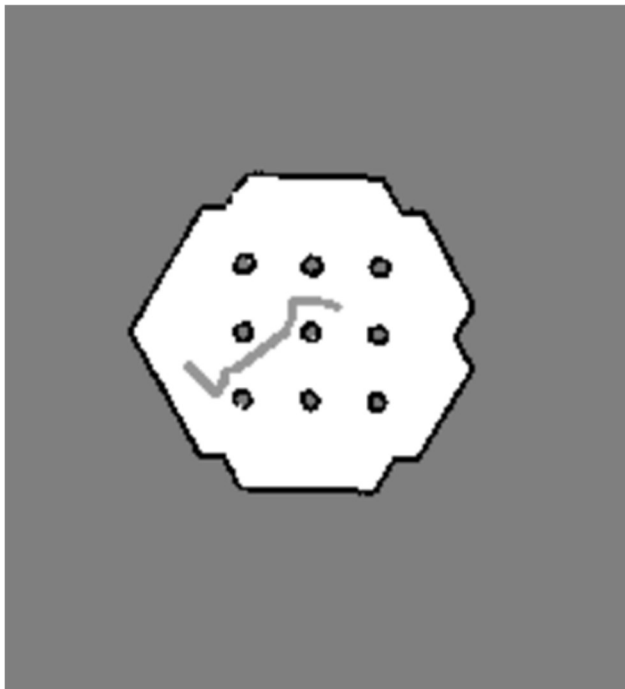
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Objective

The objective of this lab is to implement a complete motion planning system for TurtleBot3 in a simulated Gazebo environment using ROS. This involves integrating components from previous labs, including a Rapidly-exploring Random Tree (RRT) path planner, a motion planner, and a PID controller, to enable autonomous navigation from a start to a goal location.

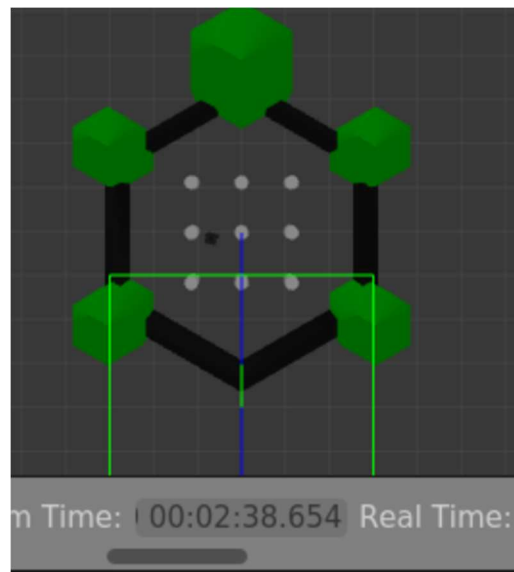
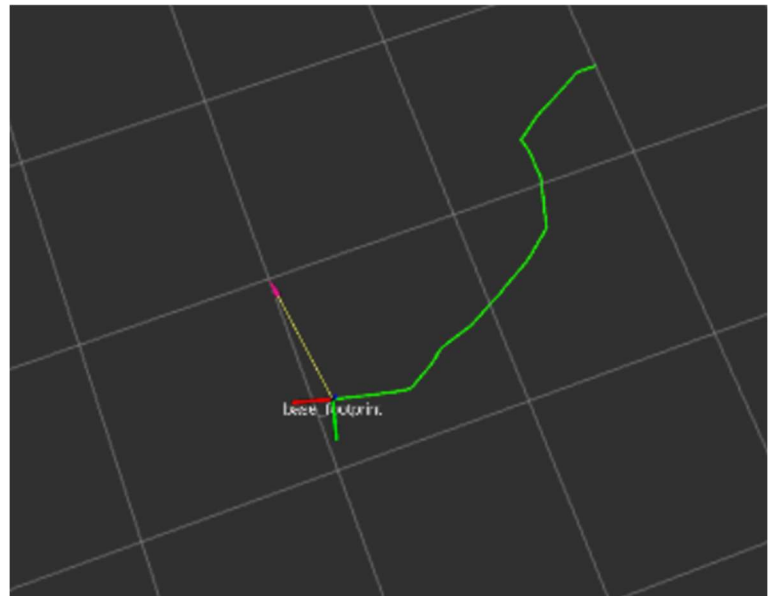
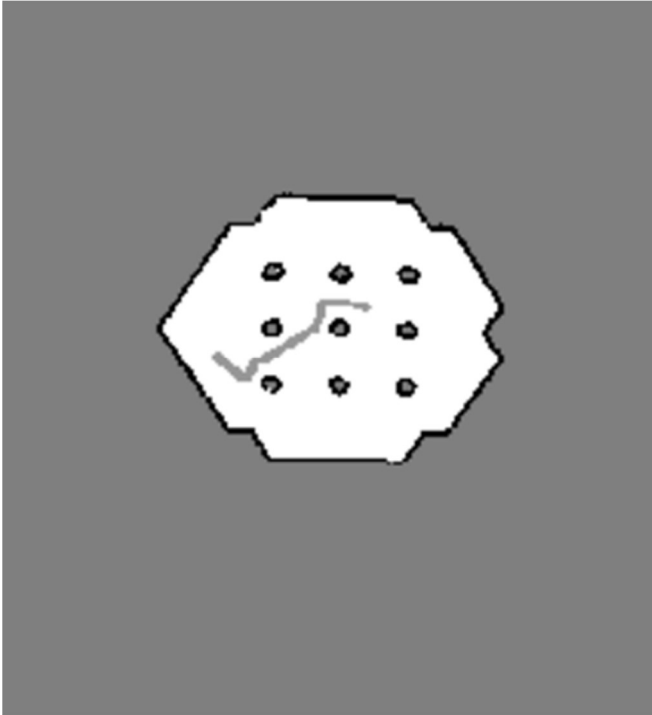
Test Point 1



Test point 1: End Goal = (0.5, 0.5)

- Start time Gazebo after running pid controller is 0:47

Test Point 2



Test point 1: End Goal = (1, 1)

- Start time Gazebo after running pid controller is 0:38